

OpenADMET PXR Structure Baseline Submission Report

Objective

This report describes the baseline structure-submission workflow used to prepare a blind-test submission package for the OpenADMET PXR structure task. The report is restricted to the completed baseline submission and is written as a methods/results summary of the submitted baseline only.

Dataset

- Input set: official blinded structure test set
- Input rows: `184`
- Input columns: `[structure, 'smiles']`
- Baseline package exists: `yes`
- Baseline package PDB count: `184`
- Baseline package size (bytes): `8349328`

Receptor Template Panel

Human holo PXR structures were curated into a monomeric receptor panel with associated reference ligands and docking boxes. Exact template-ligand duplicates were retained for curation bookkeeping, but template routing used the deduplicated selected subset.

Template panel:

- `1ILH` (PDB `1ILH`, chain `A`, ligand `SRL`, resolution `2.76` A): curated-but-not-used-for-assignment
- `1M13` (PDB `1M13`, chain `A`, ligand `HYF`, resolution `2.15` A): selected
- `1NRL` (PDB `1NRL`, chain `A`, ligand `SRL`, resolution `2.0` A): selected
- `3HVL` (PDB `3HVL`, chain `A`, ligand `SRL`, resolution `2.1` A): curated-but-not-used-for-assignment
- `3R8D` (PDB `3R8D`, chain `A`, ligand `PNU`, resolution `2.8` A): selected

Ligand Preparation And Template Assignment

Ligands were read from the official blinded structure CSV and normalised into an internal ligand identifier plus SMILES representation using the structure identifier column. For 2D inputs, protonation-state preparation was applied before docking at pH 7.4, and the dominant prepared state was used for 3D docking calculations.

Template assignment used 2D ligand similarity against template co-crystal ligands:

- fingerprint: Morgan / ECFP4-like radius 2, 2048 bits
- similarity metric: Tanimoto
- top template: always retained
- second template: retained only when within 0.10 Tanimoto of the best match and chemically non-redundant relative to the first template

Assignment summary:

- completed assignment rows: `368`
- average templates per ligand: `2.00`

Docking And Pose Refinement

Each assigned ligand-template pair was processed with the following baseline docking workflow:

1. UniDock-Pro hybrid docking in `detail` mode
2. search exhaustiveness: `8`
3. maximum poses per ligand-template pair: `3`
4. GNINA pose minimisation with `--minimize`
5. GNINA CNN rescoring with `--cnn\_scoring rescore`
6. GNINA autoboxing from the template reference ligand

Docking summary:

- completed GNINA pose rows: `973`
- average retained poses per assignment: `2.64`

Baseline Pose Selection

The blind-test baseline package was generated directly from the GNINA-minimised and CNN-rescored poses. One final complex was selected per ligand using the following ranking rule:

1. highest GNINA `CNNaffinity`
2. tie-break by most favourable GNINA `minimizedAffinity`
3. tie-break by higher ligand-template similarity
4. tie-break by lower pose rank

Each selected ligand pose was merged with the exact monomeric receptor PDB used in docking to produce one protein-ligand PDB per ligand.

Selection summary:

- baseline complexes written: `184`
- passed basic local PDB validation: `184`
- mean CNNaffinity: `6.100`
- median CNNaffinity: `6.132`
- mean minimizedAffinity: `-7.156`
- median minimizedAffinity: `-7.309`

Template usage in final baseline package:

- `1NRL`: 99 ligands
- `1M13`: 54 ligands
- `3R8D`: 31 ligands

Example selected entries:

- `x00011-1` -> template `3R8D`, pose `x00011-1\_3R8D\_pose\_001`, CNNAffinity `4.969`, minimizedAffinity `-7.103`
- `x00035-1` -> template `1NRL`, pose `x00035-1\_1NRL\_pose\_001`, CNNAffinity `5.090`, minimizedAffinity `-6.555`
- `x00046-1` -> template `1NRL`, pose `x00046-1\_1NRL\_pose\_002`, CNNAffinity `5.525`, minimizedAffinity `-6.025`
- `x00052-1` -> template `1NRL`, pose `x00052-1\_1NRL\_pose\_002`, CNNAffinity `5.616`, minimizedAffinity `-6.521`
- `x00086-1` -> template `1NRL`, pose `x00086-1\_1NRL\_pose\_003`, CNNAffinity `5.124`, minimizedAffinity `-6.013`
- `x00088-1` -> template `3R8D`, pose `x00088-1\_3R8D\_pose\_001`, CNNAffinity `4.910`, minimizedAffinity `-5.802`
- `x00113-1` -> template `1M13`, pose `x00113-1\_1M13\_pose\_003`, CNNAffinity `6.240`, minimizedAffinity `-6.887`
- `x00162-1` -> template `1NRL`, pose `x00162-1\_1NRL\_pose\_001`, CNNAffinity `6.082`, minimizedAffinity `-8.495`
- `x00186-1` -> template `1NRL`, pose `x00186-1\_1NRL\_pose\_003`, CNNAffinity `6.315`, minimizedAffinity `-7.742`
- `x00229-1` -> template `3R8D`, pose `x00229-1\_3R8D\_pose\_001`, CNNAffinity `5.273`, minimizedAffinity `-7.041`
- `x00242-1` -> template `1NRL`, pose `x00242-1\_1NRL\_pose\_001`, CNNAffinity `5.058`, minimizedAffinity `-4.105`
- `x00252-1` -> template `1NRL`, pose `x00252-1\_1NRL\_pose\_002`, CNNAffinity `5.526`, minimizedAffinity `-5.981`

Validation And Packaging

Baseline exports were subjected to local structural-format validation requiring:

- protein `ATOM` records present
- ligand `HETATM` records present
- terminal `END` record present

Validated PDB files were then bundled into the final baseline submission archive:

- archive name: `baseline\_submission\_pdb.zip`
- archive members: `184`

Reproducibility Notes

- the baseline archive contains one monomeric protein-ligand PDB per blinded ligand
- template selection, docking, and pose ranking were applied consistently across the full blinded set
- quantitative values in this report were generated directly from the saved baseline workflow outputs